



Ingestible device to monitor conditions like sleep apnea

A team of researchers from MIT, Celero Systems, and West Virginia University has introduced an ingestible capsule, roughly the size of a standard multivitamin, to offer a less intrusive approach to diagnosing conditions like sleep



Monitoring by ingestible device (Credit: MIT News)

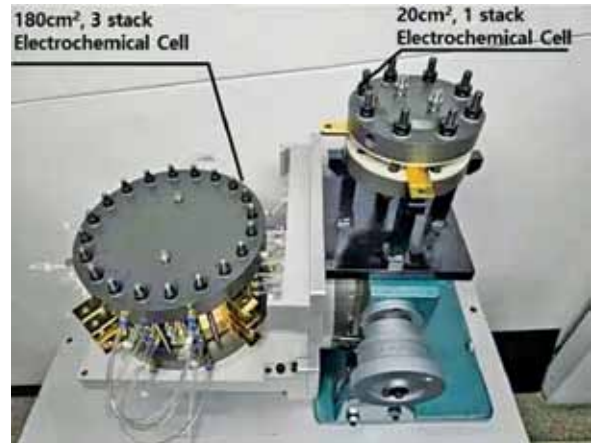
apnea. It employs an accelerometer to track breathing and heart rates, enabling the monitoring of vital signs from the patient's gastrointestinal tract. This technology also holds promise in identifying opioid overdoses among high-risk individuals. During a clinical trial, ten patients were simultaneously monitored using the ingestible capsule and traditional sleep sensors, demonstrating its precision in measuring breathing and heart rates and its effectiveness in detecting sleep apnea. The researchers envision using this sensor technology as a less invasive alternative to the current skin-based sensors for diagnosing sleep apnea. It can be used for ongoing monitoring of patients undergoing apnea treatment to ensure its efficacy.

Digital twin technology innovation for social good

The iTwin4Good Challenge, a collaborative endeavour between Enactus and Bentley Systems, encourages students to tackle UN Sustainable Development Goals using Bentley's iTwin platform. Students from 36 Brazilian universities joined forces, showcasing inventiveness, dedication to sustainability, and technical prowess. The victorious team, 'All In,' hailing from the Federal Institute of Espírito Santo Vitória, earned recognition for their AI Blood Donation Room Monitoring Project. The group designed an iTwin app with intelligent cameras and AI to oversee blood donation room occupancy, inventory management, and product demand assessment. The app effortlessly connects with external APIs, providing live updates on blood type needs. Enactus and Bentley Systems will host the iTwin World Cup in April 2024 in London, where 'All In' will compete against the winner of the UK.

Electrochemical refrigerant compressors and sustainable energy innovations

In line with the Korean government's carbon neutrality goals by 2050, researchers at the Korea Institute of Machinery and Materials (KIMM), in collaboration with Professors of Chung-Ang University, have developed an eco-friendly



Environment friendly electrochemical refrigerant compressor (Credit: www.azocleantech.com)

refrigerant compressor using electrochemistry instead of traditional mechanical methods. Unlike harmful HFC-based refrigerants, these environmentally friendly alternatives like ammonia and R1234yf have minimal environmental impact, aligning with global efforts to phase out HFCs by 2024. In contrast to mechanical ones, electrochemical compressors offer durability, no lubricant issues, and reduced noise. They improve the efficiency of heat pumps significantly. The research team has also designed an intelligent farm solution to optimise energy savings in rooftop greenhouses. This technology can handle various refrigerants, making it versatile for hydrogen infrastructure development.

Sodium-ion batteries poised to transform clean energy

Researchers at Tokyo University of Science, led by Professor Shinichi Komaba, have made a significant breakthrough in sodium-ion (Na-ion) batteries by using 'hard' carbon electrodes, significantly boosting their energy storage capacity by 1.6 times. They don't pose combustion risks, and sodium is readily available. Additionally, sodium battery production is more environmentally friendly, as it doesn't rely on rare and environmentally problematic materials like cobalt, copper, lithium, and graphite. The enhanced storage capacity of sodium-ion batteries has the potential to make electric vehicles (EVs) even more eco-friendly and cost-effective.